



Madan Mohan Malaviya Univ. of Technology, Gorakhpur

Database Management System (MCA-112) MCA 1st Sem (2020-21)



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12-03-2021 Side 1

1



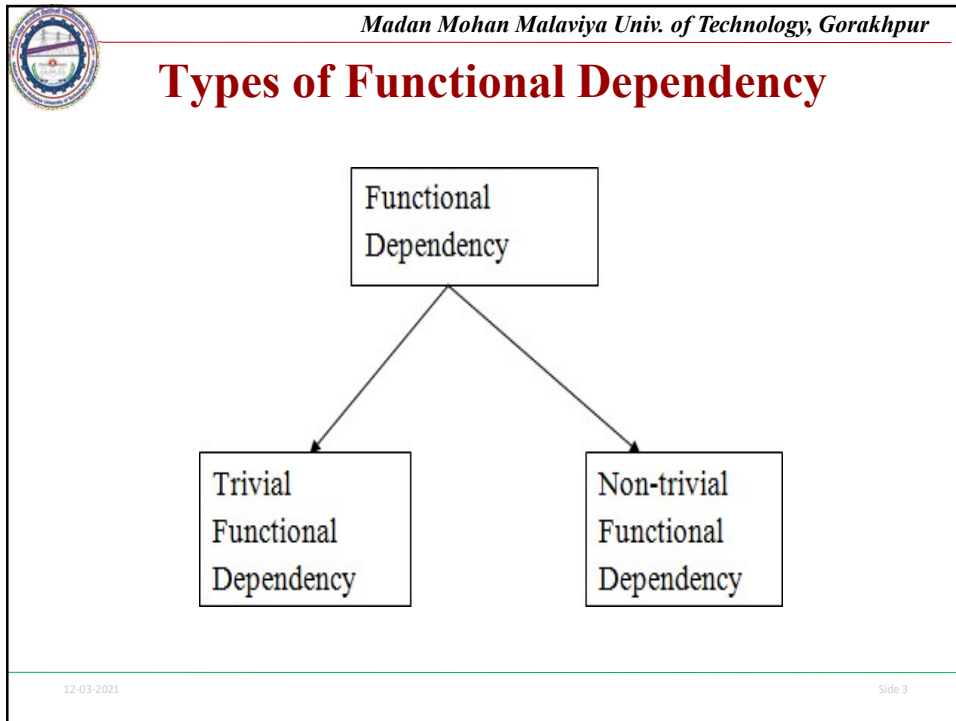
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Functional Dependency

- ❖ Functional dependency (FD) is a set of constraints between two attributes in a relation.
- ❖ Functional dependency says that if two tuples have same values for attributes $A_1, A_2, \dots, A_n$, then those two tuples must have to have same values for attributes $B_1, B_2, \dots, B_n$.
- ❖ Functional dependency is represented by an arrow sign ($\rightarrow$) that is, $X \rightarrow Y$, where X functionally determines Y . The left-hand side attributes determine the values of attributes on the right-hand side.
- ❖ Assume we have an employee table with attributes: Emp_Id, Emp_Name, Emp_Address. Here Emp_Id attribute can uniquely identify the Emp_Name attribute of employee table because if we know the Emp_Id, we can tell that employee name associated with it. Functional dependency can be written as: $\text{Emp_Id} \rightarrow \text{Emp_Name}$. We can say that Emp_Name is functionally dependent on Emp_Id.

12-03-2021 Side 2

2



3

Types of Functional Dependency

Trivial Functional Dependency

- ❖ If a functional dependency (FD) $X \rightarrow Y$ holds, where Y is a subset of X , then it is called a trivial FD. Trivial FDs always hold.
- ❖ $A \rightarrow B$ has trivial functional dependency if B is a subset of A .
- ❖ The $A \rightarrow A$, $B \rightarrow B$ dependencies are also trivial FDs.
- ❖ Consider a table with two columns `Employee_Id` and `Employee_Name`.
- ❖ $\{Employee_id, Employee_Name\} \rightarrow Employee_Id$ is a trivial functional dependency as `Employee_Id` is a subset of $\{Employee_Id, Employee_Name\}$.
- ❖ Also, $Employee_Id \rightarrow Employee_Id$ and $Employee_Name \rightarrow Employee_Name$ are trivial dependencies too.

12-03-2021 Side 4

4



Types of Functional Dependency

Non-trivial Functional Dependency

- ❖ If an FD $X \rightarrow Y$ holds, where Y is not a subset of X , then it is called a non-trivial FD.
- ❖ $A \rightarrow B$ has a non-trivial functional dependency if B is not a subset of A .
- ❖ When $A \cap B$ is NULL, then $A \rightarrow B$ is called as complete non-trivial.
- ❖ Examples: $ID \rightarrow Name$, $Name \rightarrow DOB$

12-03-2021

Side 5

5



Armstrong's Axioms

- ❖ If F is a set of functional dependencies then the closure of F , denoted as F^+ , is the set of all functional dependencies logically implied by F .
- ❖ Armstrong's Axioms are a set of rules, when applied repeatedly, generates a closure of functional dependencies.
- ❖ The Armstrong's axioms are the basic inference rule.
- ❖ Armstrong's axioms are used to conclude functional dependencies on a relational database.
- ❖ The inference rule is a type of assertion. It can apply to a set of FD(functional dependency) to derive other FD.
- ❖ Using the inference rule, we can derive additional functional dependency from the initial set.
- ❖ The Functional dependency has 6 types of inference rule:

12-03-2021

Side 6

6



Armstrong's Axioms

1. Reflexive Rule (IR_1)

In the reflexive rule, if Y is a subset of X, then X determines Y.

If $X \supseteq Y$ then $X \rightarrow Y$

Example:

$X = \{a, b, c, d, e\}$

$Y = \{a, b, c\}$ then $X \rightarrow Y$

2. Augmentation Rule (IR_2)

The augmentation is also called as a partial dependency. In augmentation, if X determines Y, then XZ determines YZ for any Z. If $X \rightarrow Y$ then $XZ \rightarrow YZ$.

Example:

For $R(ABCD)$, if $A \rightarrow B$ then $AC \rightarrow BC$

12-03-2021

Side 7

7



Armstrong's Axioms

3. Transitive Rule (IR_3)

In the transitive rule, if X determines Y and Y determines Z, then X must also determine Z.

If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$

4. Union Rule (IR_4)

Union rule says, if X determines Y and X determines Z, then X must also determine Y and Z.

If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow YZ$

Proof:

$X \rightarrow Y$ (given), $X \rightarrow Z$ (given)

$X \rightarrow XY$ (using IR_2 on 1 by augmentation with X. Where $XX = X$)

$XY \rightarrow YZ$ (using IR_2 on 2 by augmentation with Y)

$X \rightarrow YZ$ (using IR_3 on 3 and 4)

12-03-2021

Side 8

8



Armstrong's Axioms

5. Decomposition Rule (IR_5)

- ❖ Decomposition rule is also known as project rule. It is the reverse of union rule.
- ❖ This Rule says, if X determines Y and Z, then X determines Y and X determines Z separately.
- ❖ Example: If $X \rightarrow YZ$ then $X \rightarrow Y$ and $X \rightarrow Z$

Proof:

1. $X \rightarrow YZ$ (given)
2. $YZ \rightarrow Y$ (using IR_1 Rule)
3. $X \rightarrow Y$ (using IR_3 on 1 and 2)

12-03-2021

Side 9

9



Armstrong's Axioms

6. Pseudo transitive Rule (IR_6)

In Pseudo transitive Rule, if X determines Y and YZ determines W, then XZ determines W.

If $X \rightarrow Y$ and $YZ \rightarrow W$ then $XZ \rightarrow W$

Proof:

1. $X \rightarrow Y$ (given)
2. $YZ \rightarrow W$ (given)
3. $XZ \rightarrow YZ$ (using IR_2 on 1 by augmenting with Z)
4. $XZ \rightarrow W$ (using IR_3 on 3 and 2)

12-03-2021

Side 10

10



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Normalization

- ❖ **Normalization** is a database design technique that reduces data redundancy and eliminates undesirable characteristics like Insertion, Update and Deletion Anomalies.
- ❖ Normalization is used to minimize the redundancy from a relation or set of relations. It is also used to eliminate the undesirable characteristics like Insertion, Update and Deletion Anomalies.
- ❖ Normalization divides the larger table into the smaller table and links them using relationship.
- ❖ The normal form is used to reduce redundancy from the database table.

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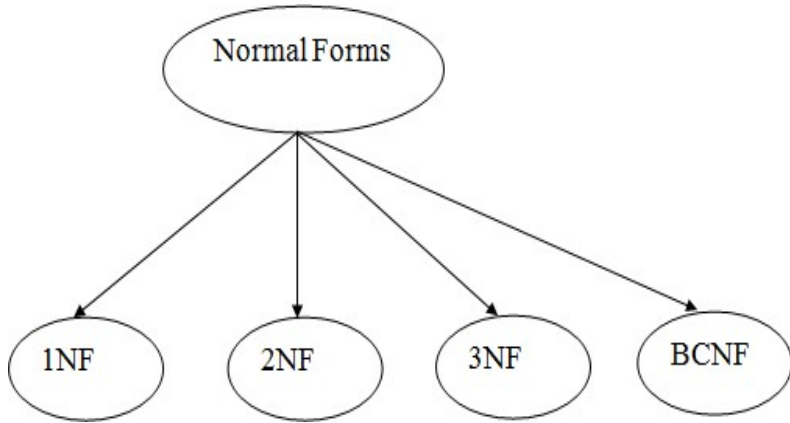
Side 11

11



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Types of Normal Form



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graph TD; A([Normal Forms]) --> B([1NF]); A --> C([2NF]); A --> D([3NF]); A --> E([BCNF]);
```

12-03-2021

Side 12

12

 <i>Madan Mohan Malaviya Univ. of Technology, Gorakhpur</i> Types of Normal Form	
Normal Form	Description
<u>1NF</u>	A relation is in 1NF if it contains an atomic value.
<u>2NF</u>	A relation will be in 2NF if it is in 1NF and all non-key attributes are fully functional dependent on the primary key.
<u>3NF</u>	A relation will be in 3NF if it is in 2NF and no transitive dependency exists.
<u>4NF</u>	A relation will be in 4NF if it is in Boyce Codd normal form and has no multi-valued dependency.
<u>5NF</u>	A relation is in 5NF if it is in 4NF and not contains any join dependency and joining should be lossless.
12-03-2021Side 13	

13

 <i>Madan Mohan Malaviya Univ. of Technology, Gorakhpur</i> Normalization	
If a database design is not perfect, it may contain anomalies, which are like a bad dream for any database administrator. Managing a database with anomalies is next to impossible.	
❖ Update anomalies – If data items are scattered and are not linked to each other properly, then it could lead to strange situations. For example, when we try to update one data item having its copies scattered over several places, a few instances get updated properly while a few others are left with old values. Such instances leave the database in an inconsistent state. ❖ Deletion anomalies – We tried to delete a record, but parts of it was left undeleted because of unawareness, the data is also saved somewhere else. ❖ Insert anomalies – We tried to insert data in a record that does not exist at all. ❖ Redundancy	
12-03-2021Side 14	

14



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First Normal Form (1NF)

- ❖ A relation will be in 1NF if it contains an atomic value.
- ❖ It states that an attribute of a table cannot hold multiple values. It must hold only single-valued attribute.
- ❖ First normal form disallows the multi-valued attribute, composite attribute, and their combinations.

12-03-2021
Slide 15

15



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Examples of 1NF

EMPLOYEE:

EMP_ID	EMP_NAME	EMP_PHONE	EMP_STATE
14	John	7272826385, 9064738238	UP
20	Harry	8574783832	Bihar
12	Sam	7390372389, 8589830302	Punjab

Relation EMPLOYEE is not in 1NF because of multi-valued attribute EMP_PHONE. The decomposition of the EMPLOYEE table into 1NF has been shown below:

EMP_ID	EMP_NAME	EMP_PHONE	EMP_STATE
14	John	7272826385	UP
14	John	9064738238	UP
20	Harry	8574783832	Bihar
12	Sam	7390372389	Punjab
	Sam	8589830302	

12-03-2021
Slide 16

16



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Examples of 1NF

FULL NAMES	PHYSICAL ADDRESS	MOVIES RENTED	SALUTATION
Janet Jones	First Street Plot No 4	Pirates of the Caribbean, Clash of the Titans	Ms.
Robert Phil	3 rd Street 34	Forgetting Sarah Marshal, Daddy's Little Girls	Mr.
Robert Phil	5 th Avenue	Clash of the Titans	Mr.

Here you see **Movies Rented** column has multiple values. **Not in 1NF.**

FULL NAMES	PHYSICAL ADDRESS	MOVIES RENTED	SALUTATION
Janet Jones	First Street Plot No 4	Pirates of the Caribbean	Ms.
Janet Jones	First Street Plot No 4	Clash of the Titans	Ms.
Robert Phil	3 rd Street 34	Forgetting Sarah Marshal	Mr.
Robert Phil	3 rd Street 34	Daddy's Little Girls	Mr.
Robert Phil	5 th Avenue	Clash of the Titans	Mr.

12-03-2021
Side 17

17



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Second Normal Form (2NF)

A relation is said to be in 2NF if it is in 1NF and all non-key attributes are fully functional dependent on the primary key i.e there is no partial dependency.

Example: Let's assume, a school can store the data of teachers and the subjects they teach. In a school, a teacher can teach more than one subject.

Teacher Table:

TEACHER_ID	SUBJECT	TEACHER_AGE
25	Chemistry	30
25	Biology	30
47	English	35
83	Math	38
83	Computer	38

In the given table, non-prime attribute TEACHER_AGE is dependent on TEACHER_ID which is a proper subset of a candidate key. That's why it violates the rule for 2NF. To convert the given table into 2NF, we decompose it into two tables:

12-03-2021
Side 18

18



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Second Normal Form (2NF)

Candidate Keys: {teacher_id, subject}, **Non prime attribute:** teacher_age

TEACHER_DETAIL table:

TEACHER_ID	TEACHER_AGE
25	30
47	35
83	38

TEACHER_SUBJECT table:

TEACHER_ID	SUBJECT
25	Chemistry
25	Biology
47	English
83	Math
83	Computer

12-03-2021
Side 19

19



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Second Normal Form (2NF)

Member Details

MEMBERSHIP ID	FULL NAMES	PHYSICAL ADDRESS	SALUTATION
1	Janet Jones	First Street Plot No 4	Ms.
2	Robert Phil	3 rd Street 34	Mr.
3	Robert Phil	5 th Avenue	Mr.

Movies Rented

MEMBERSHIP ID	MOVIES RENTED
1	Pirates of the Caribbean
1	Clash of the Titans
2	Forgetting Sarah Marshal
2	Daddy's Little Girls
3	Clash of the Titans

12-03-2021
Side 20

20



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Third Normal Form (3NF)

- ❖ A relation will be in 3NF if it is in 2NF and not contain any transitive dependency.
- ❖ 3NF is used to reduce the data duplication. It is also used to achieve the data integrity.
- ❖ If there is no transitive dependency for non-prime attributes, then the relation must be in third normal form.
- ❖ A relation is in third normal form if it holds at least one of the following conditions for every non-trivial function dependency $X \rightarrow Y$.
 - ✓ X is a super key.
 - ✓ Y is a prime attribute, i.e., each element of Y is part of some candidate key.

Example

EMPLOYEE_DETAIL table:

EMP_ID	EMP_NAME	EMP_ZIP	EMP_STATE	EMP_CITY
222	Harry	201010	UP	Noida
333	Stephan	02228	US	Boston
444	Lan	60007	US	Chicago
555	Katharine	06389	UK	Norwich
	John	462007	MP	

21



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Third Normal Form (3NF)

Super key in the table above:

1. {EMP_ID}, {EMP_ID, EMP_NAME}, {EMP_ID, EMP_NAME, EMP_ZIP}so on

Candidate key: {EMP_ID}

Non-prime attributes: In the given table, all attributes except EMP_ID are non-prime.

Here, EMP_STATE & EMP_CITY dependent on EMP_ZIP and EMP_ZIP dependent on EMP_ID. The non-prime attributes (EMP_STATE, EMP_CITY) transitively dependent on super key(EMP_ID). It violates the rule of third normal form.

That's why we need to move the EMP_CITY and EMP_STATE to the new <EMPLOYEE_ZIP> table, with EMP_ZIP as a Primary key.

12-03-2021
Slide 22

22

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Third Normal Form (3NF)

EMPLOYEE table:

EMP_ID	EMP_NAME	EMP_ZIP
222	Harry	201010
333	Stephan	02228
444	Lan	60007
555	Katharine	06389
666	John	462007

EMPLOYEE_Zip table:

EMP_ZIP	EMP_STATE	EMP_CITY
201010	UP	Noida
02228	US	Boston
60007	US	Chicago
06389	UK	Norwich
462007	MP	Bhopal

23

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Third Normal Form (3NF)

Suppose a company wants to store the complete address of each employee, they create a table named employee_details that looks like this:

emp_id	emp_name	emp_zip	emp_state	emp_city	emp_district
1001	John	282005	UP	Agra	Dayal Bagh
1002	Ajeet	222007	TN	Chennai	M-City
1006	Lora	282007	TN	Chennai	Urrapakkam
1101	Lilly	292008	UK	Pauri	Bhagwan
1201	Steve	222999	MP	Gwalior	Ratan

12-03-2021
Side 24

24



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Third Normal Form (3NF)

Super keys: {emp_id}, {emp_id, emp_name}, {emp_id, emp_name, emp_zip}...so on

Candidate Keys: {emp_id}

Non-prime attributes: all attributes except emp_id are non-prime as they are not part of any candidate keys.

Here, emp_state, emp_city & emp_district dependent on emp_zip. And, emp_zip is dependent on emp_id that makes non-prime attributes (emp_state, emp_city & emp_district) transitively dependent on super key (emp_id). This violates the rule of 3NF. To make this table complies with 3NF we have to break the table into two tables to remove the transitive dependency:

12-03-2021
Side 25

25



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Third Normal Form (3NF)

Employee table:

emp_id	emp_name	emp_zip
1001	John	282005
1002	Ajeet	222008
1006	Lora	282007
1101	Lilly	292008
1201	Steve	222999

Employee_Zip table:

emp_zip	emp_state	emp_city	emp_district
282005	UP	Agra	Dayal Bagh
222008	TN	Chennai	M-City
282007	TN	Chennai	Urrapakkam
292008	UK	Pauri	Bhagwan
222999	MP	Gwalior	Ratan

12-03-2021
Side 26

26



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Boyce Codd normal form (BCNF)

- ❖ BCNF is the advance version of 3NF. It is stricter than 3NF.
- ❖ A table is in BCNF if every functional dependency $X \rightarrow Y$, X is the super key of the table.
- ❖ For BCNF, the table should be in 3NF, and for every FD, LHS is super key.

Example: Let's assume there is a company where employees work in more than one department.

EMPLOYEE table:

EMP_ID	EMP_COUNTRY	EMP_DEPT	DEPT_TYPE	EMP_DEPT_NO
264	India	Designing	D394	283
264	India	Testing	D394	300
364	UK	Stores	D283	232
364	UK	Developing	D283	549

12-03-2021
Slide 27

27



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Boyce Codd normal form (BCNF)

In the above table Functional dependencies are as follows:

1. $EMP_ID \rightarrow EMP_COUNTRY$
2. $EMP_DEPT \rightarrow \{DEPT_TYPE, EMP_DEPT_NO\}$

Candidate key: {EMP-ID, EMP-DEPT}

The table is not in BCNF because neither EMP_DEPT nor EMP_ID alone are keys.

To convert the given table into BCNF, we decompose it into three tables:

EMP_COUNTRY table:

EMP_ID	EMP_COUNTRY
264	India
264	India

12-03-2021
Slide 28

28



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Boyce Codd normal form (BCNF)

EMP_DEPT table:

EMP_DEPT	DEPT_TYPE	EMP_DEPT_NO
Designing	D394	283
Testing	D394	300
Stores	D283	232
Developing	D283	549

EMP_DEPT_MAPPING table:

EMP_ID	EMP_DEPT
D394	283
D394	300
D283	232
D283	549

12-03-2021
Side 29

29



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Boyce Codd normal form (BCNF)

Functional dependencies:

1. EMP_ID → EMP_COUNTRY
2. EMP_DEPT → {DEPT_TYPE, EMP_DEPT_NO}

Candidate keys:

For the first table: EMP_ID

For the second table: EMP_DEPT

For the third table: {EMP_ID, EMP_DEPT}

Now, this is in BCNF because left side part of both the functional dependencies is a key.

12-03-2021
Side 30

30



Boyce Codd normal form (BCNF)

Example 2:

Consider a relation given below:

SP (Sno, Sname, Pno, Qty), Here Sname is considered unique for each Sno. So, FD of above relation is:

$(Sno, Pno) \rightarrow Qty, (Sname, Pno) \rightarrow Qty, Sno \rightarrow Sname$

$Sname \rightarrow Sno$. Is this relation in 3NF? Check for it to be in BCNF?

How can you solve this problem?

Solution: There are two overlapping candidate keys.

(Sno, Pno) and $(Sname, Pno)$ ie

$(Pno, Sno) \rightarrow (Qty, Sname)$ and $(Pno, Sname) \rightarrow (Qty, Sno)$

Both the relations are in 3NF because every nonkey attribute is nontransitively fully functional dependent on the primary key.

12-03-2021

Side 31

31



Boyce Codd normal form (BCNF)

SP is not in BCNF because it has four determinants:

(Sno, Pno) , $(Sname, Pno)$, Sno , $Sname$. Out of these four determinants, two determinants (Sno, Pno) and $(Sname, Pno)$ are unique but Sno and $Sname$ determinants are not candidate keys.

Now to make this relation in BCNF, we have to non-loss decompose this relation into two projections:

SN (Sno, Sname)

SPP (Sno, Pno, Qty).

Here, **SN** relation has two determinants, and both are unique.

SPP has one determinant (Sno, Pno) and is also unique.

Thus, all the anomalies are removed.

12-03-2021

Side 32

32



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Fourth normal form (4NF)

- ❖ A relation will be in 4NF if it is in Boyce Codd normal form and has no multi-valued dependency.
- ❖ For a dependency $A \twoheadrightarrow B$, if for a single value of A, multiple values of B exists, then the relation will be a multi-valued dependency.

STUDENT

STU_ID	COURSE	HOBBY
21	Computer	Dancing
21	Math	Singing
34	Chemistry	Dancing
74	Biology	Cricket
59	Physics	Hockey

- ❖ The given STUDENT table is in 3NF, but the COURSE and HOBBY are two independent entity. Hence, there is no relationship between COURSE and HOBBY.
- ❖ In the STUDENT relation, a student with STU_ID, **21** contains two courses, **Computer** and **Math** and two hobbies, **Dancing** and **Singing**. So, there is a Multi-valued dependency on STU_ID, which leads to unnecessary repetition of data.

12-03-2021
Side 33

33



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Fourth normal form (4NF)

So, to make the above table into 4NF, we can decompose it into two tables:

STUDENT_COURSE

STU_ID	COURSE
21	Computer
21	Math
34	Chemistry
74	Biology
59	Physics

STUDENT_HOBBY

STU_ID	HOBBY
21	Dancing
21	Singing
34	Dancing
74	Cricket
59	Hockey

12-03-2021
Side 34

34



Fifth normal form (5NF)

- ❖ A relation is in 5NF if it is in 4NF and not contains any join dependency and joining should be lossless.
- ❖ 5NF is satisfied when all the tables are broken into as many tables as possible in order to avoid redundancy.
- ❖ 5NF is also known as Project-join normal form (PJ/NF).

Example

SUBJECT	LECTURER	SEMESTER
Computer	Anshika	Semester 1
Computer	John	Semester 1
Math	John	Semester 1
Math	Akash	Semester 2
Chemistry	Praveen	Semester 1

12-03-2021

Side 35

35



Fifth normal form (5NF)

- ❖ In the above table, John takes both Computer and Math class for Semester 1 but he doesn't take Math class for Semester 2. In this case, combination of all these fields required to identify a valid data.
- ❖ Suppose we add a new Semester as Semester 3 but do not know about the subject and who will be taking that subject, so we leave Lecturer and Subject as NULL. But all three columns together acts as a primary key, so we can't leave other two columns blank.
- ❖ So, to make the above table into 5NF, we can decompose it into three relations P1, P2 & P3:

P1

SEMESTER	SUBJECT
Semester 1	Computer
Semester 1	Math
Semester 1	Chemistry
Semester 2	Math

12-03-2021

Side 36

36



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Fifth normal form (5NF)

P2

SUBJECT	LECTURER
Computer	Anshika
Computer	John
Math	John
Math	Akash
Chemistry	Praveen

P3

SEMSTER	LECTURER
Semester 1	Anshika
Semester 1	John
Semester 1	John
Semester 2	Akash
Semester 1	Praveen

12-03-2021
Side 37

37

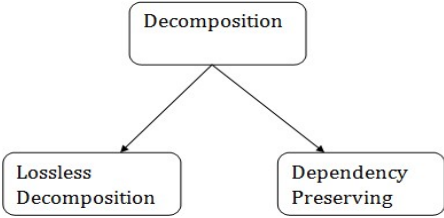


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Relational Decomposition

- ❖ When a relation in the relational model is not in appropriate normal form then the decomposition of a relation is required.
- ❖ In a database, it breaks the table into multiple tables.
- ❖ If the relation has no proper decomposition, then it may lead to problems like loss of information.
- ❖ Decomposition is used to eliminate some of the problems of bad design like anomalies, inconsistencies, and redundancy.

Types of Decomposition



```

graph TD
    A[Decomposition] --> B[Lossless Decomposition]
    A --> C[Dependency Preserving]
  
```

12-03-2021
Side 38

38



Lossless Decomposition

- ❖ If the information is not lost from the relation that is decomposed, then the decomposition will be lossless.
- ❖ The lossless decomposition guarantees that the join of relations will result in the same relation as it was decomposed.
- ❖ The relation is said to be lossless decomposition if natural joins of all the decomposition give the original relation.

EMPLOYEE_DEPARTMENT table

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY	DEPT_ID	DEPT_NAME
22	Denim	28	Mumbai	827	Sales
33	Alina	25	Delhi	438	Marketing
46	Stephan	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production
60	Jack	40	Noida	678	Testing

12-03-2021

Side 39

39



Lossless Decomposition

The above relation is decomposed into two relations EMPLOYEE & DEPARTMENT.

EMPLOYEE table

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY
22	Denim	28	Mumbai
33	Alina	25	Delhi
46	Stephan	30	Bangalore
52	Katherine	36	Mumbai
60	Jack	40	Noida

DEPARTMENT table

DEPT_ID	EMP_ID	DEPT_NAME
827	22	Sales
438	33	Marketing
869	46	Finance
575	52	Production
678	60	Testing

12-03-2021

Side 40

40



Lossless Decomposition

Now, when these two relations are joined on the common column "EMP_ID", then the resultant relation will look like:

Employee ⋈ Department

EMP_ID	EMP_NAME	EMP_AGE	EMP_CITY	DEPT_ID	DEPT_NAME
22	Denim	28	Mumbai	827	Sales
33	Alina	25	Delhi	438	Marketing
46	Stephan	30	Bangalore	869	Finance
52	Katherine	36	Mumbai	575	Production
60	Jack	40	Noida	678	Testing

Hence, the decomposition is Lossless join decomposition.

12-03-2021

Side 41

41



Dependency Preserving

- ❖ It is an important constraint of the database.
- ❖ In the dependency preservation, at least one decomposed table must satisfy every dependency.
- ❖ If a relation R is decomposed into relation R1 and R2, then the dependencies of R either must be a part of R1 or R2 or must be derivable from the combination of functional dependencies of R1 and R2.
- ❖ For example, suppose there is a relation R (A, B, C, D) with functional dependency set (A→BC). The relation R is decomposed into R1(ABC) and R2(AD) which is dependency preserving because FD A→BC is a part of relation R1(ABC).

12-03-2021

Side 42

42



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Multivalued Dependency

- ❖ Multivalued dependency occurs when two attributes in a table are independent of each other but, both depend on a third attribute.
- ❖ A multivalued dependency consists of at least two attributes that are dependent on a third attribute that's why it always requires at least three attributes. Suppose there is a bike manufacturer company which produces two colours (white and black) of each model every year.

BIKE_MODEL	MANUF_YEAR	COLOR
M2011	2008	White
M2001	2008	Black
M3001	2013	White
M3001	2013	Black
M4006	2017	White
M4006	2017	Black

12-03-2021
Side 43

43



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Multivalued Dependency

- ❖ Here columns COLOR and MANUF_YEAR are dependent on BIKE_MODEL and independent of each other.
- ❖ In this case, these two columns can be called as multivalued dependent on BIKE_MODEL. The representation of these dependencies is shown below:

$$\text{BIKE_MODEL} \twoheadrightarrow \text{MANUF_YEAR}$$

$$\text{BIKE_MODEL} \twoheadrightarrow \text{COLOR}$$

- ❖ This can be read as "BIKE_MODEL multidetermined MANUF_YEAR" and "BIKE_MODEL multidetermined COLOR".

12-03-2021
Side 44

44



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Join Dependency

- ❖ Join decomposition is a further generalization of Multivalued dependencies.
- ❖ If the join of R1 and R2 over C is equal to relation R, then we can say that a join dependency (JD) exists.
- ❖ Where R1 and R2 are the decompositions R1(A, B, C) and R2(C, D) of a given relations R (A, B, C, D).
- ❖ Alternatively, R1 and R2 are a lossless decomposition of R.
- ❖ A JD $\bowtie \{R_1, R_2, \dots, R_n\}$ is said to hold over a relation R if R1, R2, ..., Rn is a lossless-join decomposition.
- ❖ The $*(A, B, C, D), (C, D)$ will be a JD of R if the join of join's attribute is equal to the relation R.
- ❖ Here, $*(R_1, R_2, R_3)$ is used to indicate that relation R1, R2, R3 and so on are a JD of R.

12-03-2021

Side 45

45



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Inclusion Dependency

- ❖ Multivalued dependency and join dependency can be used to guide database design although they both are less common than functional dependencies.
- ❖ Inclusion dependencies are quite common. They typically show little influence on designing of the database.
- ❖ The inclusion dependency is a statement in which some columns of a relation are contained in other columns.
- ❖ The example of inclusion dependency is a foreign key. In one relation, the referring relation is contained in the primary key column(s) of the referenced relation.
- ❖ Suppose we have two relations R and S which was obtained by translating two entity sets such that every R entity is also an S entity.
- ❖ Inclusion dependency would be happen if projecting R on its key attributes yields a relation that is contained in the relation obtained by projecting S on its key attributes.
- ❖ In inclusion dependency, we should not split groups of attributes that participate in an inclusion dependency.
- ❖ In practice, most inclusion dependencies are key-based that is involved only keys.

12-03-2021

Side 46

46



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Canonical Cover

- ❖ In the case of updating the database, the responsibility of the system is to check whether the existing functional dependencies are getting violated during the process of updating.
- ❖ In case of a violation of functional dependencies in the new database state, the rollback of the system must take place.
- ❖ A canonical cover or irreducible a set of functional dependencies FD is a simplified set of FD that has a similar closure as the original set FD.

12-03-2021 Side 47

47



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12-03-2021 Side 48

48